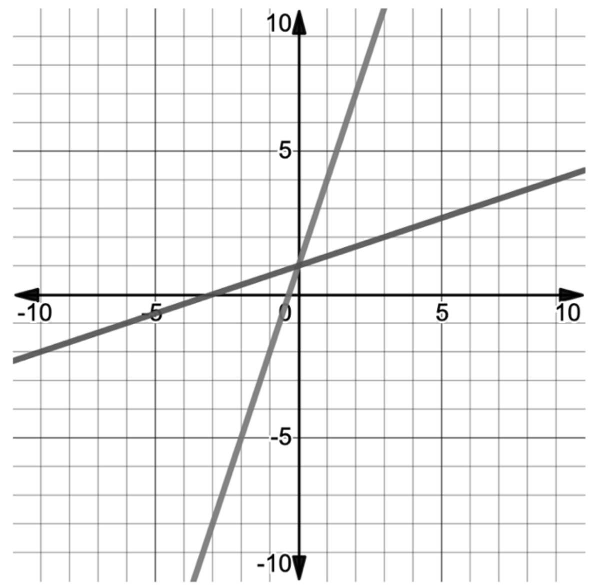
Algebra 1
Unit 4
Lesson 9 Practice

Name Answer Key___________
Date____________________
Period__________________

Use the following information to answer questions 1 through 5.

The lines of the equations $y - 7 = 3(x - 2)$ and $-\frac{1}{3}x + y = 1$ are shown on the same coordinate grid.




1. What is the slope of the line representing the equation $y - 7 = 3(x - 2)$?

   Slope= 3 $\quad y - y_1 = m(x - x_1)$

   $m$ =slope

2. What is the slope of the line representing the equation $-\frac{1}{3}x + y = 1$?

   Slope= $\frac{1}{3}$ $\quad +\frac{1}{3x} \quad +\frac{1}{3}x$

   $y = \frac{1}{3}x + 1$

   $y = mx + b$

   $m$ =slope

3. What is the slope of a line that is perpendicular to $-\frac{1}{3}x + y = 1$?

   Perpendicular will be opposite sign and reciprocal.

   $y = \frac{1}{3}x + 1$

   $\frac{1}{3} \perp \; -3$

   $= -3$

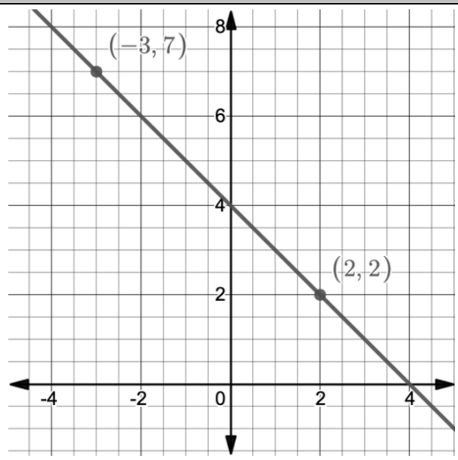
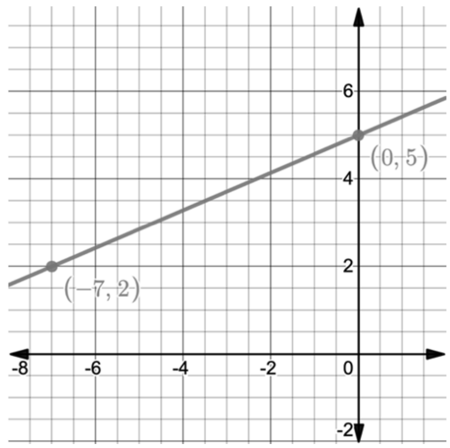
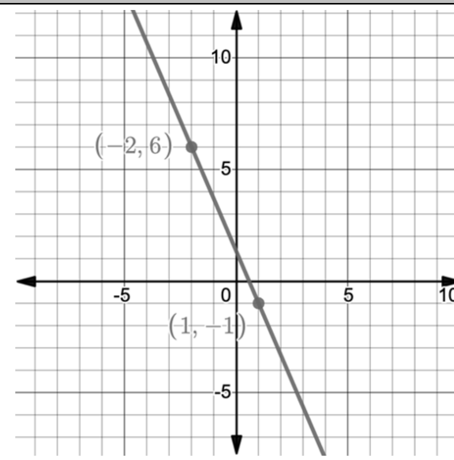
4. Explain how to determine if two lines are perpendicular.

   Find their slopes and compare them. They should be opposites and reciprocals.

5. Determine if the lines are parallel, perpendicular, or neither.

   The lines are neither. $\frac{1}{3}$ vs 3 are neither parallel nor perpendicular.

Use the following lines to answer questions 6 through 8.

| **Line $D$** | **Line $G$** | **Line $K$** |
|---|---|---|
|   |   |   |
| **Line $M$** | **Line $N$** | **Line $P$** |
| $-x - y = 6$ | $y = -x + 8$ | $\frac{7}{3}x - y = -1$ |

6. Which lines are parallel? $\frac{y_2 - y_1}{x_2 - x_1}$ or $\frac{\text{rise}}{\text{run}}$ parallel = same slope

Slopes: $D: -1$ $M: -1$
$G: \frac{3}{7}$ $N: -1$
$K: \frac{-7}{3}$ $P: \frac{7}{3}$

$-x - y = 6$
$-y = x + 6$
$y = -x - 6$

$\frac{7}{3}x - y = -1$
$-y = \frac{-7}{3}x - 1$
$y = \frac{7}{3}x + 1$

Same slope = $D, M$ and $N$

7. Which lines are perpendicular?
$\frac{3}{7}$ and $-\frac{7}{3}$ opposite sign reciprocals = $G$ and $K$

8. What is the slope of a line that is parallel to Line $P$?
Line $P = \frac{7}{3}$ Parallel to $P = \frac{7}{3}$

9. Determine if the lines of the two equations are parallel, perpendicular, or neither.

| **Equation $H$** | **Equation $D$** |
|---|---|
| $-2x - 4y = -8$ | $-2x + 4y = -8$ |

$H: -4y = 2x - 8$
$y = -\frac{2}{4}x + 2$
$m = \frac{-1}{2}$

$D: \frac{4y}{4} = \frac{2x - 8}{4}$
$y = \frac{1}{2}x - 2$
$m = \frac{1}{2}$

$-\frac{1}{2}$ vs $\frac{1}{2}$ = neither

10. Determine if the lines of the two equations are parallel, perpendicular, or neither.

| **Equation $S$** | **Equation $W$** |
|---|---|
| $-5x + y = -3$ | $y - 8 = 5(x - 3)$ |

$S$: $y = 5x - 3$

$m = 5$

$W$: $m = 5$

Same slope = parallel